

Express JS Exercise Instructions

Overview

Build an Express.js application and submit your work as a ZIP file and a GitHub repository. Include screenshots of your API calls (e.g., Postman) showing outputs or any errors. Also add static middleware so that an `instruction.html` file can be served from a `public` folder.

Submission naming

ZIP: StudentID_COMP3123-exec04.zip

Repository: StudentID_COMP3123-exec04

Replace *StudentID* with your actual ID

Required Deliverables

- Create an Express JS application named `StudentID_COMP3123-exec04`.
- Add the Express framework and nodemon to `package.json`.
- Implement the routes described below.
- Add static middleware to serve `instruction.html` from the `public` folder.
- Upload screenshots showing output or any errors (e.g., from Postman).
- Submit your GitHub repository link and upload the ZIP: `StudentID_COMP3123-exec04.zip`.

Endpoints to Implement

GET /hello

Returns `Hello Express JS` as plain text.

GET /user (query parameters)

Accepts query string parameters: `?firstname=<value>&lastname=<value>`.

- If parameters are not provided, default to `firstname = "Pritesh"` and `lastname = "Patel"`.

Example request: `GET /user?firstname=John&lastname=Doe`

Example response:

```
{ "firstname": "John", "lastname": "Doe" }
```

POST /user/:firstname/:lastname (path parameters)

Expects path parameters for `firstname` and `lastname`.

Example request: `POST /user/John/Doe`

Example response:

```
{ "firstname": "John", "lastname": "Doe" }
```

POST /users (body – array of users)

Expects a JSON array in the request body, each item with `firstname` and `lastname`.

Example request body:

```
[
  { "firstname": "Pritesh", "lastname": "Patel" },
  { "firstname": "John", "lastname": "Doe" },
  { "firstname": "John", "lastname": "Rome" }
]
```

Example response:

```
[
  { "firstname": "Pritesh", "lastname": "Patel" },
  { "firstname": "John", "lastname": "Doe" },
  { "firstname": "John", "lastname": "Rome" }
]
```

Static Middleware

Use `express.static()` to serve the provided `instruction.html` (download the attached file for this exercise) from a `public` folder.

Example project structure

```
StudentID_COMP3123-exec04/
├─ package.json
├─ index.js
├─ public/
  └─ instruction.html  <- place the attached file here
```

index.js snippet

```
const express = require("express");
const app = express();

app.use(express.json());

// ✅ Serve static files in /public (instruction.html will be at /instruction.html)
app.use(express.static("public"));

// GET /hello
app.get("/hello", (req, res) => {
```

```
res.type("text/plain").send("Hello Express JS");
});

// GET /user?firstname=&lastname=
app.get("/user", (req, res) => {
  const firstname = req.query.firstname || "Pritesh";
  const lastname = req.query.lastname || "Patel";
  res.json({ firstname, lastname });
});

// POST /user/:firstname/:lastname
app.post("/user/:firstname/:lastname", (req, res) => {
  const { firstname, lastname } = req.params;
  res.json({ firstname, lastname });
});

// POST /users (expects an array of { firstname, lastname })
app.post("/users", (req, res) => {
  const users = Array.isArray(req.body) ? req.body : [];
  res.json(users);
});

const PORT = process.env.PORT || 3000;
app.listen(PORT, () => console.log(`Server running on http://localhost:${PORT}`));
```

Project Setup (Guidance)

1) Initialize project

```
mkdir StudentID_COMP3123-exec04
cd StudentID_COMP3123-exec04
npm init -y
npm install express
npm install --save-dev nodemon
mkdir public
# Place your instruction.html file into the ./public folder
```

2) package.json scripts

```
{
  "name": "StudentID_COMP3123-exec04",
  "version": "1.0.0",
  "scripts": {
    "start": "node index.js",
    "dev": "nodemon index.js"
  },
  "dependencies": {
```

```
"express": "^4.19.0"
},
"devDependencies": {
  "nodemon": "^3.0.0"
}
}
```

3) Run and test

```
# Development (auto-restart):
npm run dev

# Production-style start:
npm start

# Test endpoints (examples):
curl http://localhost:3000/hello
curl "http://localhost:3000/user?firstname=John&lastname=Doe"
curl -X POST http://localhost:3000/user/John/Doe
curl -X POST http://localhost:3000/users \
  -H "Content-Type: application/json" \
  -d '[{"firstname":"Pritesh","lastname":"Patel"}, {"firstname":"John","lastname":"Doe"}]'
# Static file (after placing it in public/):
# http://localhost:3000/instruction.html
```

What to Submit

- ☐ GitHub repository created: StudentID_COMP3123-exec04 and link submitted.
- ☐ ZIP uploaded: StudentID_COMP3123-exec04.zip.
- ☐ Screenshots of Postman (or similar) showing each endpoint response and any errors.
- ☐ Static file served from /public/instruction.html and reachable at /instruction.html.